Baud rate: Baud rate is the rate at which the number of signal elements or changes to the signal occurs per second when it passes through a transmission medium.

Data Transmission rate or simply Transmission rate: The data transmission rate is the volume of data ***transmitted*** over a transmission channel or via a data interface within a specified unit of time. The units used for this are baud or bits/s (bps).

Now, in the book or on the slides, you would find

- transmission rate: *bandwidth*
- bandwidth (bits per second)

and,

- transmits packet into access network at ***transmission rate R***
  - **link transmission rate**, aka **link *capacity*, *aka link bandwidth***

In the book's context, you can use these terms **synonymously**:

- Transmission rate
- Link transmission rate
- Data Transmission rate
- Link bandwidth
- Link capacity

However, the actual **Bandwidth** is defined as:

Bandwidth: The maximum amount of data **transmitted** over an internet connection or link in a given amount of time.

Transmission rate or at a particular time, *link bandwidth* can vary over time but given a bandwidth of a link, that remains the same. For instance, your link has **30 bps** bandwidth but when transmitting data, it may not be always 30 bps but rather go down at times!

On the other hand,

Throughput: is the rate (bits/time unit) at which bits ***transferred*** between sender and receiver.

- **instantaneous:** rate at given point in time
- **average:** rate over longer period of time

Throughput is basically talking about the volume of data that has actually been **transferred** from the sender to the receiver. There is a difference between the terms: **transmission** and **transfer**. A sender transmits, i.e., pushes the bits to the channel or link or path but unless it is received on the other end, the transfer has not been completed. **Clear?**